

Checksum Algorithm

Credit card numbers use the following algorithm to check if the number is valid.

Given a credit card number of 16 digits:
"4578423013769219"

Use the first 15 digits:
{4, 5, 7, 8, 4, 2, 3, 0, 1, 3, 7, 6, 9, 2, 1}

Starting with the first digit (Index 0 in an Array), multiply every second digit by 2. This would give the following:

{ (4*2), 5, (7*2), 8, (4*2), 2, (3*2), 0, (1*2), 3, (7*2), 6, (9*2), 2, (1*2) }

This gives

{8, 5, 14, 8, 8, 2, 6, 0, 2, 3, 14, 6, 18, 2, 2}

Every time you have a two-digit number in the sequence add the two digits together to create a one-digit result:

{8, 5, (1+4), 8, 8, 2, 6, 0, 2, 3, (1+4), 6, (1+8), 2, 2}

This gives

{8, 5, 5, 8, 8, 2, 6, 0, 2, 3, 5, 6, 9, 2, 2}

Finally, add all the numbers together:

$8 + 5 + 5 + 8 + 8 + 2 + 6 + 0 + 2 + 3 + 5 + 6 + 9 + 2 + 2 = 71$

Add the total to the 16th digit (Index 15 in an Array) (known as the check digit)

The result must be an even multiple of 10. In this case:

$71 + 9 = 80$

Create a Main Class and a CheckSum Class

In the CheckSum class add the following method:

```
public static boolean checkSum(String accNo) {  
  
}
```